

ANAESTHESIA — PERIOPERATIVE COMPLICATIONS

Safety & risk

- Anaesthesia among **safest specialties**. Incidence: **minor 1:5, major 1:100, death 1:100,000** (~25/million developed countries).
- Risk factors**: patient (ASA status, comorbidities, age), procedure (extent, urgency), environmental (staffing, equipment).
- Mortality trend declining (pre-1970 3.6:10,000 → after 1990 0.3:10,000).
- Respiratory = most frequent** complication; cardiac/renal/neuro less frequent but high impact.
- Patient most feared: vomiting, gagging on tube, pain, nausea, **recall (awareness)**, weakness. Clinician priorities: prevent death, recall, nerve/corneal injury.

Airway complications

- Difficult airway**: pre-op assessment critical; plan + backup; avoid multiple attempts → **CICO (cannot intubate, cannot oxygenate)**. Manage with SAD/LMA; **cricothyroidotomy** for CICO emergency. **Capnograph = best monitor to avoid fatal unrecognised oesophageal intubation**.
- Pulmonary aspiration**: common in emergencies; silent → ARDS. **Prevention: fasting (6–8h solids, 2h liquids), NG tube, antacids, prokinetics (metoclopramide), RSI, awake extubation**.
- Upper airway obstruction**: signs = absent bag movement, high airway pressure, flat capnograph. Causes: tongue fall, foreign body, trauma. **Management: triple manoeuvre (head tilt + chin lift + jaw thrust) + adjuncts (oropharyngeal/nasopharyngeal/LMA/ETT) + suction**.
- Lower airway obstruction**: bronchospasm, aspiration, pneumothorax → hypoxia, wheeze, ↑airway pressure.
- Postop pulmonary complications**: infection, atelectasis, respiratory failure. Risk: age, ASA ≥2, CHF, abdominal/thoracic surgery. Prevent: minimally invasive, avoid GA, minimal opioids, early mobilization, physio.
- OSA**: difficult airway, hypoxia, delayed extubation → avoid GA, ↓opioids, CPAP/ICU.
- Airway trauma & fire**: lips/teeth/trachea (perforation = high mortality); barotrauma/pneumothorax; diathermy/laser fire risk.

Cardiovascular complications

- MI**: O₂ supply/demand imbalance (hypoxia, anaemia, hypo/hypertension, tachycardia). Optimize cardiac status; delay elective if unstable.
- Hypotension**: ↓CO/↓PVR; hypovolaemia, anaesthetic drugs, high spinal block → cerebral/renal hypoperfusion, arrest.
- Hypertension/arrhythmias**: anxiety/pain (pre-op), pain/full bladder/hypercapnia (post-op); arrhythmias usually benign — treat cause.
- Stroke**: emboli (AF), thrombosis, neck rotation, post-bypass. Vascular access complications: infection, air embolism, pneumothorax, haematoma.
- Thromboembolism (DVT/PE)**: age, BMI, immobility, surgery, HRT/OCP. **Gas embolism** (0.5–1 mL/kg significant): cyanosis, hypotension, tachycardia, engorged neck veins, arrest → CPR + CVP line aspiration.

Neurological & other

- **Awareness:** unintentional (faulty equipment, empty vaporizer), pharmacodynamic variability, or intentional (sick patient).
- **Delayed recovery:** sedatives, muscle relaxants, opioids, thiopental/propofol, volatile agents, surgical/metabolic.
- **CNB/PNB complications:** CNB — infection, abscess, headache, nerve injury; PNB — **LAST (local anaesthetic systemic toxicity)**, haematoma. Use US guidance.
- Delirium (elderly), convulsions (drug/hypoxia/toxicity), **shivering** (after volatile even if normothermic).
- Renal/hepatic: rare direct; common with pre-existing disease → avoid nephro/hepatotoxic drugs, maintain perfusion.
- **Drug errors** (wrong drug/dose/route/timing); **PONV** (multifactorial) → prevent with TIVA, antiemetics, regional (avoid GA).

High-yield one-liners

- **Respiratory = most frequent complication; capnograph = best monitor (oesophageal intubation).**
- **CICO → cricothyroidotomy.** Aspiration prevention = fasting + RSI.
- **Upper airway obstruction → triple manoeuvre (head tilt + chin lift + jaw thrust).**
- **Awareness = empty vaporizer/faulty equipment; LAST = peripheral nerve block complication.**
- **Gas embolism = engorged neck veins + cyanosis → CPR + CVP aspiration.**

ANAESTHESIA — PERIOPERATIVE COMPLICATIONS (Revision Table)

Bold = high-yield. Dr Gona Ahmed.

OVERVIEW & AIRWAY	
Topic	Key points
Incidence	Minor 1:5, major 1:100, death 1:100,000 ; respiratory most frequent
Risk factors	Patient (ASA, comorbidity, age), procedure (extent/urgency), environment
Difficult airway	Pre-op assess; avoid multiple attempts → CICO → cricothyroidotomy; capnograph = best monitor (oesophageal intubation)
Aspiration	Emergencies; → ARDS. Prevent: fasting 6–8h solid/2h liquid, RSI, NG, prokinetics
Upper airway obstruction	Absent bag movement, ↑pressure, flat capnograph (tongue/FB/trauma) → triple manoeuvre + adjuncts
Postop pulmonary	Atelectasis/infection; risk age/ASA≥2/CHF; prevent early mobilization, ↓opioids
OSA	Difficult airway, hypoxia → avoid GA, CPAP, ICU
CARDIOVASCULAR & NEUROLOGICAL	
Topic	Key points
MI	O2 supply/demand imbalance; optimize, delay elective if unstable
Hypotension	↓CO/PVR; hypovolaemia, drugs, high spinal → cerebral/renal hypoperfusion
Stroke/vascular access	Emboli (AF), neck rotation; line complications (pneumothorax, air embolism)
Thromboembolism	DVT/PE (age, BMI, immobility, OCP). Gas embolism: engorged neck veins + cyanosis → CPR + CVP aspiration
Awareness	Faulty equipment (empty vaporizer), variability, intentional
CNB/PNB	CNB: abscess, headache, nerve injury; PNB: LAST ; use US
Other	Delirium (elderly), shivering (post-volatile), PONV → TIVA/antiemetics/regional ; drug errors